

Shuhan Wang

MSc Machine Learning | World Models / Model-Based RL

Available full-time, on-site in Hong Kong for 12 months from 14 Sep 2026

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PROFILE

UCL MSc Machine Learning candidate researching learned world models, model-based RL and experimental benchmark design. Led CLIMB, extending DreamerV3 with prediction-guided action selection in Craftax-Symbolic and imagined actor-critic rollouts. Seeking a World Models RA role to extend this experience towards visual and embodied settings, with an interest in world-model-based policy learning and exploration.

EDUCATION

University College London (UCL) — MSc Machine Learning

Sep 2025 – Dec 2026 (expected)

Current taught average: 79.3/100; taught component completed

University of Manchester — BSc Mathematics and Physics

Sep 2022 – Jun 2025

First-Class Honours; overall average: 76.8/100

RESEARCH EXPERIENCE

CLIMB: DreamerV3 for Open-Ended Continual Learning [🔗](#)

Feb 2026 – Apr 2026

UCL group research; project lead and first author

- **World-model-guided policy learning:** Investigated how world-model predictions can guide feasible actions in **Craftax-Symbolic**. Extended DreamerV3 in JAX / Flax with an RSSM-based action-context head; used its predictions with hand-specified feasibility rules in both environment interaction and imagined actor-critic rollouts.
- **Exploration and continual learning:** Combined soft feasibility biases with episodic spatial / inventory novelty rewards and novelty-learnability-uniform replay; used component ablations to examine their contributions to exploration and achievement progression.
- **Controlled evaluation:** Evaluated 19 configurations across three seeds. At 10M steps, mean achievement rate over the final 200 episodes was 0.200 ± 0.005 versus 0.099 ± 0.008 for the matched 50:50 recent/uniform replay baseline (mean $\pm$ SD); ablations identified soft masking as the main isolated contributor.

OpenQFT-Bench: Towards Open-Ended Learning

Jul 2026 – present

Independent research; in development

- **Benchmark design:** Developing a benchmark using simplified particle-physics worlds as a controlled testbed for learning through experimentation. Agents choose collision experiments, infer hidden particles and interaction rules, and revise world models from partial, noisy observations.
- **Learning to learn across worlds:** Designed an evaluation protocol with different underlying physical laws across generated universes. The long-term objective is to test whether agents acquire transferable discovery strategies that accelerate learning in unfamiliar worlds.
- **Implemented foundation:** Built the JAX simulator, experimental environment and structural / predictive evaluation stack, with Sequential Monte Carlo and expected-information-gain components for hypothesis updating and experiment selection.

RopeBand: RoPE-Compatible Q/K Rotations [🔗](#)

May 2026 – present

First author; under review at TMLR

- **Mathematical analysis and empirical validation:** Derived closed-form rotations for a scoped variance surrogate and evaluated dynamic 4-bit quantisation across multiple checkpoints and seeds; documented higher perplexity than full-head Hadamard for the evaluated pairwise-only transforms and analysed the surrogate-quantiser mismatch.

SWE-Router: Data and Evaluation for Agent Routing

Apr 2026 – Jul 2026

Co-author; ICML 2026 DL4C Workshop

- **Reproducible data, evaluation and training:** Independently owned multi-model trajectory collection, per-step cost annotation and official SWE-bench / SWE-Smith grading using Docker / HPC Apptainer and SGE. Ran Qwen2.5-Coder-7B LoRA value-model training with Hugging Face Accelerate on $3 \times$ A100 80GB GPUs.

TECHNICAL SKILLS

World models / RL: DreamerV3, RSSM, latent imagination, actor-critic, intrinsic exploration, structured replay

Modelling / evaluation: Bayesian inference, Sequential Monte Carlo, experimental design, multi-seed ablations, bootstrap, partial observability

Implementation: Python, JAX / Flax, PyTorch, Hugging Face Accelerate, Git, Linux, Docker / Apptainer, SGE / HPC